

Vedran Mitic

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PROFESSIONAL SUMMARY

Master's student in Artificial Intelligence and Machine Learning with a strong background in scalable data systems and distributed processing. Experienced in Python, Apache Spark, Hadoop, and containerized environments (Docker). Passionate about building cloud-based data analytics solutions and implementing ML-driven architectures within AWS ecosystems.

EDUCATION

Faculty of Electronic Engineering, University of Nis

Nis, Serbia

Master's Degree in Artificial Intelligence and Machine Learning

Nov 2025 – Present

Faculty of Electronic Engineering, University of Nis

Nis, Serbia

Bachelor of Electrical Engineering and Computer Science (GPA: 8.22/10)

Oct 2021 – Oct 2025

Universidad Politécnica de Madrid

Madrid, Spain

Erasmus+ Student Mobility – ICT

Feb 2025 – Jul 2025

TECHNICAL SKILLS

Programming: Python, SQL, C/C++, MATLAB

Machine Learning & Data: Supervised Learning, Unsupervised Learning, Data Preprocessing, Neural Networks, Scikit-learn, Pandas, NumPy, Explainable AI (XAI)

Big Data: Apache Spark, Hadoop (HDFS)

Tools: Docker, Podman, Git, Jupyter Notebook, PyCharm, VS Code

Systems: PLC Programming, SCADA Systems, Embedded Systems (Raspberry Pi)

SELECTED PROJECTS

- **Data Quality Engineering in Financial Predictive Modeling** GitHub
Analyzed the impact of Data Quality on ML model performance by designing a preprocessing and validation pipeline for bank customer churn prediction. Combined theoretical research with practical implementation in Python (Pandas, Scikit-learn, H2O Auto ML) to evaluate reliability and model stability.
- **Big Data Infrastructure with Spark & Hadoop** GitHub
Designed and deployed a distributed data processing architecture using Docker, Hadoop (HDFS), and Apache Spark to process 4GB+ datasets, simulating scalable infrastructure for large-scale analytics workloads.
- **Strategic Transaction Segmentation & Profitability Analysis** GitHub
Developed an unsupervised ML pipeline to segment 9,000+ transactions and identify margin leakage patterns, translating raw data into actionable business insights.
- **Explainable Machine Learning on Biological Data** GitHub
Developed classification and regression models for biological datasets and applied SHAP and LIME to interpret model behavior and improve transparency.
- **ANN Architecture Optimization for Data Classification (Bachelor Thesis)** GitHub
Conducted empirical analysis of three-layer neural network architectures using MATLAB, comparing SCG and Levenberg–Marquardt optimization algorithms for binary and multi-class classification.

CERTIFICATIONS

- Red Hat OpenShift Development I: Introduction to Containers with Podman (DO188) – Red Hat, 2026

Languages: Serbian (Native), English (C1), Spanish (A2)